

Paper 1 (JINST) — proposed outline

For approval before writing. Draft 3, 9 October 2026

Title

An algorithm-independent lower limit on combinatorial fake tracks in high-occupancy tracking detectors

Target: JINST regular article, about 20 pages, 7–8 figures. **Readers:** physicists who know track reconstruction at large collider experiments; no tutorial material on fits or Kalman filters.

The message in three sentences

1. When noise hits become dense enough compared with the detector resolution, some random combinations of hits fit a track as well as real tracks do; no algorithm that keeps real tracks at the same efficiency can reject them, because the information is not in the data.
2. Their expected number is exactly $E = P \times \prod n_i$ (P : probability that one random combination passes the χ^2 cut; n_i : noise hits per plane), and P , which can be 10^{-26} , is computed with an exact rescaling of the detector plus ordinary Monte Carlo.
3. E is therefore a lower limit on the fake rate of any algorithm; we show how it scales with hit density and resolution and apply it to a six-layer barrel tower with Muon Collider-like backgrounds.

Sections

1. Introduction (1.5 pp)

- High-occupancy tracking: HL-LHC, heavy ions, the Muon Collider beam-induced background. Fake rates are normally obtained by full simulation of one specific algorithm.
- The question asked here: at what density does tracking become impossible *for any algorithm*?
- Precursor: Casarsa, Jindariani, Ristori, JINST 20 (2025) P04030, Sec. 8.5 (inflated-resolution Monte Carlo scaled back with a product-of-resolutions law). This paper makes that argument exact and general.

- Related ideas outside tracking, cited and contrasted: random alignments of points (Kendall & Kendall; Broadbent), the “number of false alarms” of a contrario detection (Desolneux, Moisan, Morel), rare-event methods that inflate a scale and extrapolate (Bucher; Naess et al.). Accidental coincidences in counter systems and fake roads in associative-memory triggers as the familiar product-of-occupancies heuristics.
- Plan of the paper.

2. Definitions and the lower-limit statement (2 pp)

- Detector model: N planes, noise hits uniformly distributed with density ρ_i on plane i , measurement resolution σ (per coordinate, per plane).
- Track hypothesis and acceptance: least-squares fit, $\chi^2 < \text{cut}$, with the cut fixed by the efficiency for genuine tracks (one-sided 3σ : $P(\chi^2 > \text{cut}) = 1.35 \times 10^{-3}$).
- The counting identity $E = P \coprod n_i$: exact for the mean by linearity of expectation, whatever the correlations between overlapping combinations.
- **The lower-limit statement** (wording as agreed): any algorithm that uses only the hit positions and keeps genuine tracks passing this χ^2 cut must also accept, on average, at least E pure-noise combinations. State the class of algorithms it covers; what can lower it (an algorithm that drops candidates through hit sharing or ambiguity resolution also loses real tracks) and what it does not count (partly-fake tracks, which only add to it).

2.x Requiring at least k of N planes

- **Acceptance rule:** a candidate is accepted if its hits on some set S of at least k planes pass the χ^2 cut for that set’s ndof; each ndof has its own cut, at the same efficiency for genuine tracks (one-sided 3σ). For 5 of 6: ndof = 8 for the full set, 6 for each 5-plane subset.
- **Expected number of accepted combinations**, by the same linearity argument: $E(\geq k \text{ of } N) = \sum P_S \prod_{i \in S} n_i$, the sum running over the sets S of at least k planes — for 5 of 6, E_6 plus six terms $E_5(S)$. Each P_S needs its own calibration, since removing a plane changes the geometry.
- **From combinations to distinct fake tracks:** the sum counts some fakes more than once — a 6-hit fake usually contains 5-hit subsets that also pass, and two 5-hit fakes can share 4 hits, which any algorithm would merge. Hence: largest single term $\leq$ distinct fakes $\leq$ sum.
- **Why the overlaps are small in practice:** E_5 is expected to exceed E_6 by a large factor, and two 5-hit fakes sharing 4 hits need a second hit inside an already tiny window. The sum is then a good estimate, and the lower-limit statement rests on it with the bounds stated.
- **Checked, not just argued:** the end-to-end validation (Section 4) counts distinct fakes after merging candidates that share hits, and compares them with the sum and the bounds.

3. The probability that one random combination passes (4 pp)

Developed throughout on the $B = 0$ straight line, the simplest case, which shows the whole method; Section 5 adds the field.

- **Whitened coordinates from the start:** every measured coordinate divided by its resolution. The fit becomes an unweighted least-squares fit, χ^2 is the squared length of the residual vector, and $\chi^2 < c$ means that this vector lies inside a **sphere** of radius $\sqrt{c}$ in the ndof-dimensional residual space. Planes of different size and different resolutions need no separate treatment: in these coordinates each plane is simply a box of side W_i/σ_i .
- **The small- χ^2 power law, derived for readers who have not met it** (about one page):
 - With m measured coordinates and p fitted parameters, the fitted residuals r are the projection of the whitened measurements x onto the $(m - p) = \text{ndof}$ -dimensional space orthogonal to the track model; $\chi^2 = |r|^2$.
 - $P(\chi^2 < c)$ is the probability that r falls inside the sphere $|r| < \sqrt{c}$ in that space: $P = \int_{|r| < \sqrt{c}} f_r(r) d^{\text{ndof}}r$.
 - When $\sqrt{c}$ is small compared with the scale over which f_r varies (the whitened plane sizes W_i/σ_i), f_r is constant over the sphere, so $P \approx f_r(0) V_{\text{ndof}} c^{\text{ndof}/2}$, with $V_d = \pi^{d/2}/\Gamma(d/2 + 1)$ the volume of the unit sphere: hence $K = f_r(0) V_{\text{ndof}}$, and the exponent $\text{ndof}/2$ follows from geometry alone.
 - $f_r(0)$ is the density of combinations that lie exactly on a track. For a linear model it can be written as an integral over the track parameters of the hit density along the exact tracks (with a constant Jacobian), which gives an optional analytic cross-check of K in the straight-line case.
 - The same power appears in the familiar small-argument form of the χ^2 distribution, $P \approx (c/2)^{\text{ndof}/2}/\Gamma(\text{ndof}/2 + 1)$ for Gaussian residuals (incomplete-gamma series; cite DLMF §8.7 or Abramowitz & Stegun) — there with a different constant, because here the hits are uniform noise, not Gaussian around a track. For straight lines it reproduces the $(w/L)^{k-2}$ law for chance alignments of k random points (Kendall & Kendall 1980; Broadbent 1980). Small-ball probabilities (Li & Shao 2001) as the general mathematical setting.
- Exact rescaling: in whitened coordinates shrinking the planes by λ (or, equivalently, enlarging every σ by λ) multiplies χ^2 by λ^2 exactly, since the residuals are linear in the data; so $P_{\text{true}}(c) = P_{\text{shrunk}}(c/\lambda^2)$, and the rare event becomes common in the shrunken detector.
- Calibration: ordinary Monte Carlo on the shrunken detector; fit K on the plateau of $P/c^{\text{ndof}/2}$ with the exponent fixed; how to check the plateau; choice of λ (smallest plane 5–20 resolutions wide).
- Consequence: $P \propto \sigma^{\text{ndof}}$, the resolution scaling, is the same symmetry read the other way.
- Relation to rare-event methods and to extreme-value theory: here the scaling is an exact symmetry and the exponent is known, so only K is fitted.

4. Validation (2.5 pp)

- The plateau of $P/c^{\text{ndof}/2}$ over several decades of c , for the straight line and the helix (figure).
- The resolution law checked numerically (the $10^{\text{ndof}/N}$ ratio of crossing densities).
- **New:** an end-to-end test in small, tractable detectors where fakes are frequent enough to count directly: generate noise-only events, fit *all* combinations, count those passing the cut, and compare with $P \prod n_i$ — for the straight line and for the helix, for 6 of 6 and k of N (figure).

5. Tracks in a solenoidal field (3.5 pp)

- Tracks from the beam line: the fit splits into a bending view (circle through the beam line, parameters b, κ) and a depth view (a line), $\text{ndof} = 2N - 4$.
- The exact helix fit (Levenberg-Marquardt, several starting points; one sentence on why a single start would underestimate fakes).
- Scale equivariance of the circle model: the rescaling stays exact provided the curvature limit scales with λ ($\kappa \rightarrow \lambda\kappa$).
- Curvature acceptance (the p_T threshold): the accepted fraction grows linearly with κ_{max} and does not depend on σ .
- Comparison with the $B = 0$ line of Section 3: what the field costs and gains (one more fitted parameter, but a curvature acceptance).

6. Worked example: a six-layer barrel tower (3 pp)

- Geometry and densities *inspired by* a Muon Collider tracker (citations): six layers at radii ~ 160 – 1500 mm, projective tower with 1 m^2 outermost plane, densities after background-suppression cuts of 10^{-4} – 10^{-2} hits/ mm^2 , all rounded to two significant figures; $\sigma = 50 \text{ }\mu\text{m}$; $B = 5 \text{ T}$; $p_T > 10 \text{ GeV}/c$.
- Results: P for the exact helix; E for 6 of 6 and for 5 of 6 (six subsets, compared).
- Headroom: E vs a common density factor ($\propto \mu^6$) and vs resolution ($\propto \sigma^8$), the factor at which one fake is reached.

7. Uncertainties and limitations (1.5 pp)

- Statistical uncertainty of K ; systematic from the plateau range; fit convergence.
- Assumptions: Gaussian resolution without tails, multiple scattering neglected (high- p_T tracks), projective idealization, noise uniform within a plane (local density is what enters).
- What would raise the true fake rate above the limit: partly-fake tracks, correlated noise (clusters, loopers).

8. Conclusions (0.5 pp)

Appendices

- A. Notation.
- B. Calibration constants (K, ndof, λ , statistics) for every configuration used.

- C. Code and data availability (git repository; archived version).
- Acknowledgements, including the AI disclosure statement required by IOP.

What is left out of the compendium

The history of the work and the “pitfalls” narratives (one sentence each where useful); the discussion on choosing λ (one rule of thumb); the conservative quadratic model; the angular-acceptance and impact-parameter cuts; the first-order validity study (one remark); everything that refers to simulation data, which goes to paper 2.

New work needed before writing results

1. Recalibrate on the rounded geometry: exact helix for 6 of 6 and for each of the six 5-plane subsets (the slow runs; time to be estimated first).
2. Uncertainties: statistical (bootstrap) and plateau-range systematic for every K.
3. The end-to-end validation of Section 4 (new Monte Carlo, small detectors; straight line and helix; 6 of 6 and k of N).
4. All figures regenerated by scripts in docs/papers/jinst_fake_rate/.

Decided (9 October 2026)

1. Title: the one above.
2. Whitened coordinates from the start; spheres, not ellipsoids.
3. Only the exact helix fit; the $B = 0$ line as the pedagogical case. No conservative quadratic model.
4. No angular-acceptance or impact-parameter cuts.
5. The end-to-end validation in small, tractable detectors is included.
6. k of N gets its own subsection (2.x), with the counting of distinct fakes checked by the end-to-end validation.
7. The small- χ^2 power law is derived in the paper, with references, not assumed known.